677644

COMPUTER NETWORKS

CO1: Understand the concept of OSI and TCP/IP reference model
CO2: Understand the basics of data transmission at physical layer.
CO3: Understand the channel allocation using ALOHA, CSMA and CSMA/CD.
CO4: Apply error detection and correction technique to eliminate transmission error.
CO5: Analyze the fixed and variable length address (IPv4) subneting for the given scenarios.
CO6: Understand the design issues of the transport layer.
CO7: Understand the mechanism of protocols at application layer such as FTP, HTTP, Telnet, DNS.
CO8: Understand IPv6 addressing and differentiate it from IPv4.

Printed Pages: 3 **University Roll No. .....**

Mid Term Examination, Odd Semester 2024-25
B.Tech.(CSE), 2 Year, III Semester
BCSC 0008 & Computer Networks

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:
1. Attempt all questions
2. Do not switch between section A and Section B; first, solve one section, then move to second
3. Use only a blue pen to answer
4. No B sheet will be given

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	How Manchester Line Coding Scheme is differing from Differential Manchester Line Coding Scheme. Consider a bit stream 10110010 and draw the Manchester and Differential Manchester coding?	3	2	An	C
2	A small start-up company is experiencing network connectivity issues within their office based on OSI layers. They suspect that the problem lies within the OSI model layers. As a network technician, how would you approach troubleshooting this issue considering different services of OSI layer?	3	1	R	F
3	A school needs to set up a network for 30 computers in a classroom. The budget is limited, and cost-effectiveness is a priority. The network should support file sharing and internet access. What network topology would you choose to minimize costs while still	3	2	U	F

	maintaining basic functionality, and what are the trade-offs of your choice?				
4	A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system (all stations together) produces 250 frames per second.	3	3	A	P
5	Assume that 12-bit hamming code word consist of 8-bit data and 4 check bits is $d_8d_7d_6d_5c_8d_4d_3d_2c_4d_1c_2c_1$ where the data bits and the check bits are given in the following tables: Data bits d_8 d_7 d_6 d_5 d_4 d_3 d_2 d_1 1 1 0 x 0 1 0 1 Check bits c_8 c_4 c_2 c_1 y 0 1 0 Calculate the correct values of x and y?	3	4	A	M

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Consider the binary code that consists of only four valid code words as given below: 00000, 01011, 10101, 11110 Let the minimum Hamming distance of the code p and the maximum number of erroneous bits that can be corrected by the code be q. Then the values of p and q are? Explain Cyclic Redundancy Check (CRC). A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is x^3+1. What is the actual bit string transmitted? Suppose that the third bit from the left is inverted during transmission. Show that this error is detected at the receiver's end.	5	4	A	M

7	Datagram	Path Length	Visited Switches	5	3	A	P
	1	3200 km	1,3,5				
	2	11,700 km	1,2,5				
	3	12,200 km	1,2,3,5				
	4	10,200 km	1,4,5				
	5	10,700 Km	1,4,3,5				
	We assume that the delay for each switch (including waiting and processing) is 3,10, 20, 7, and 20 ms respectively. Assuming that the propagation speed is 2 x 10^8 m/s, find the order of arrival at the destination and the delay for each. Ignore any other delays in transmission.						
8	A and B are the only two stations on Ethernet. Each has a steady queue of frames to send. Both A and B attempts to transmit a frame, collide and A wins first back off race. At the end of this successful transmission by A, both A and B attempt to transmit and collide. What is the probability that A, B wins the second back off race?			5	3	A	M